Nirvaan Kohli

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Education

- ***West Morris Mendham High School*** ***GPA: 3.98/4.00*** *2025–Present*
  - 5 AP courses in 2 years (maximum available); Coursework: AP Computer Science Principles, AP Statistics, AP Precalculus, Algebra II

Technical Skills

Languages: Python, JavaScript, TypeScript, C/C++, Swift, HTML/CSS **Frameworks/Tools:** React, Electron, Express, SwiftUI, Flask, SQLite, Git, MapKit, PROS
Data & ML: PyTorch, scikit-learn, Pandas, NumPy | Agent Tools: LangChain, LangGraph, MCP, OpenAI API, Codex

Experience & Research

- ***AP CSP Teaching Assistant / Web Developer West Morris Mendham HS*** *2026–Present*
  - Helped the AP Computer Science Principles teacher develop a new course website for lessons, classroom activities, and student learning resources; assist with classroom instruction.
- ***Student Researcher — Activation Steering (SGAS)Algoverse Research*** *2026–Present*
  - Developed transformer inference and hidden-state capture pipelines for controlled steering experiments; contributed to model generation and evaluation across safety datasets.
  - Own the CAST comparison baseline: reproducing published results and evaluating condition vectors and gating scores against success-guided steering; research ongoing and unpublished.
- ***Hackathon Organizer — Haven*** ***1435 Capital, Princeton, NJ*** *Nov 14–15, 2026*
  - Organizing a two-day student hackathon (Nov. 14–15) for 50+ expected participants; raised $4,000+ in prize value and coordinate outreach, sponsors, and event logistics.
- ***Head Programmer*** ***VEX Robotics Team 42747R*** *2025–Present*
  - Developed C++ autonomous navigation and localization algorithms using Extended Kalman Filters (EKF) and Monte Carlo localization with PROS/EZ-Template; competed at VEX States.
- ***Programming Instructor*** ***Oakland Gurudwara*** *2025–Present*
  - Teach coding fundamentals to 10 elementary students and guide 7 advanced students through programming projects.
- ***Hackathon Organizer*** ***Fanpage YSWS / Hack Club*** *2025*
  - Ran a two-week coding event with 16 submissions, 251 recorded programming hours, and over $754 in prizes.

Selected Projects

- ***Echo*** *— Python, macOS, iMessage, MCP, Codex*
  - Developed an always-on macOS background agent with an iMessage bridge, persistent task execution, scheduling, connected tools, and optional Codex worker integration.
- ***OmniLab*** *— VEX Robotics, Path Planning*
  - Built a programming toolkit for VEX robotics teams with path planning, simulation, and collaborative development tools.
- ***MORPH.00 — Adaptive Keyboard*** *— Embedded Systems, Machine Learning*
  - Designed a customizable keyboard with ML-assisted, context-aware key mappings that change based on the application or activity in use.
- ***Argument Framework Engine*** *— Python, Streamlit, Graph Visualization*
  - Built an application that converts Reddit discussions into interactive argument graphs, linking claims by supporting and opposing relationships.

Awards & Activities

Big Red Hacks — Winner | **Ridge Hacks** — 3rd Place | **HackMCST X** — 1st Place (Middle School) | FBLA State Qualifier (IT) | VEX State Championship